



Regulation: R22

Code: 22EC205/1

II B.Tech. I Semester Regular Examinations – December 2022

SIGNALS & SYSTEMS

(ECE)

Time: 150 Min

Max. Marks: 80M

SECTION – A

Answer all Four questions

4×10M=40M

1. A periodic impulse train $\delta_T(t)$ with period T is used for collecting samples from a continuous time signal $x(t)$.
 - (a) What mathematical equation represents $\delta_T(t)$.
 - (b) Is it possible to find the Fourier transform? If so, what is the Fourier transform of $\delta_T(t)$?
 - (c) If $x(\omega)$ is shown in below Figure (1).

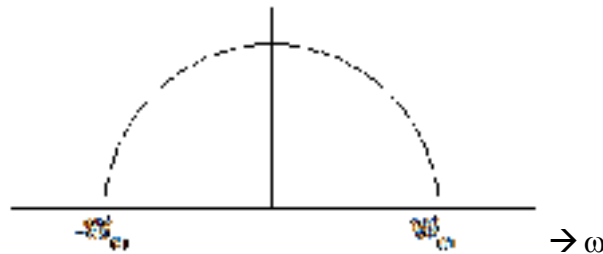


Figure (1)

What will be the spectrum of sampled signal? Sketch it and what mathematical equations justify your answer?

2. If the transfer function of system is shown in below Figure (2).

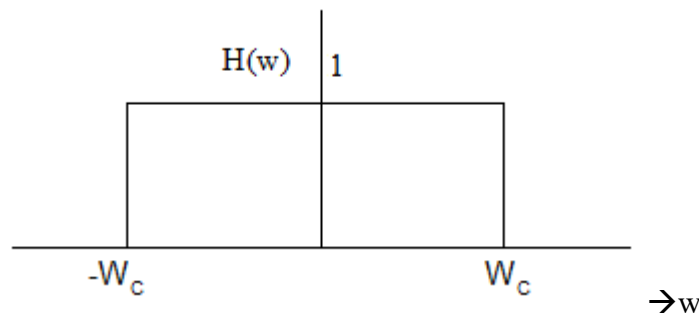


Figure (2)

Analytically find the $h(t)$. Is it possible to implement in real time? Justify your answer if Yes or No. Comment on the system in terms of causality.

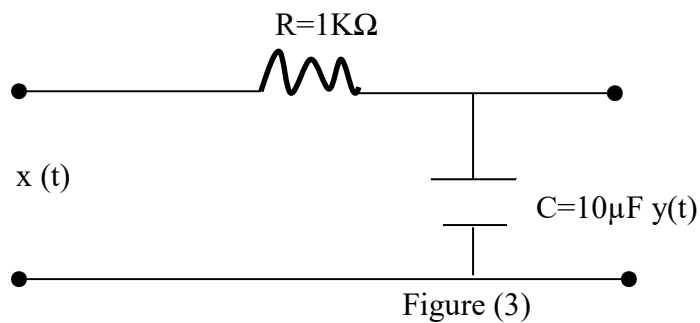
3. Consider a continuous time LTI system for which the input $x(t)$ and output $y(t)$ are related by the differential equation

$$\frac{d^2 y(t)}{dt^2} - \frac{d y(t)}{dt} - 2 y(t) = x(t)$$

Let $X(s)$ and $Y(s)$ denote Laplace transforms of $x(t)$ and $y(t)$, respectively, and let $H(s)$ denote the Laplace transform of $h(t)$, the system impulse response.

- (a) Determine $H(s)$ as a ratio of two polynomials in s . Sketch the pole-zero pattern of $H(s)$.
- (b) Determine $h(t)$ for each of the following cases:
 1. The system is stable
 2. The system is causal
 3. The system is neither stable nor causal.

4. Consider the following LTI system shown in Figure (3).



- (a) What is the order of the system?
- (b) Based on the frequency response, comment on this RC network, what type of filter it is?
- (c) What is the impulse response of the system?
- (d) Find Step response of the system?
- (e) If a square wave of fundamental frequency 100Hz was applied at input, then output $y(t)$ consists of which frequencies?

SECTION - B

Answer all Two Questions:

2 X 20M=40 M

5. Consider the following LTI system shown in Figure (4).

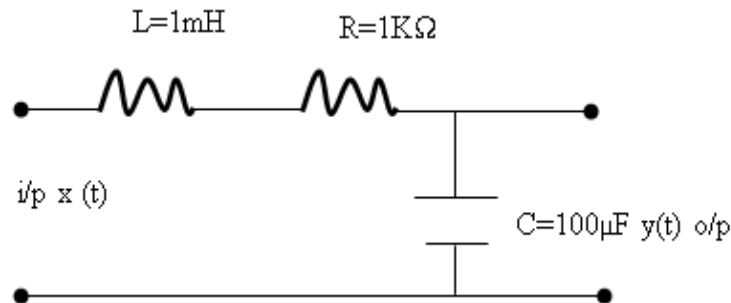


Figure (4)

- (a) Represent the system by using mathematical equation and mention the order of system.
 - (b) Can we find impulse response and step response for this system if so what is its responses?
 - (c) Plot the pole zero map and comment on the stability?
 - (d) Based on the frequency response, comment on the system, what type of filter it is?
 - (e) Based on impulse response, comment on the system with respect to causality and stability?
6. Let $x(t)$ is a continuous time, strictly time limited signal.
- (a) If impulse-train sampling is performed on $x(t)$, what is the required sampling frequency? Is it possible to recover the signal properly? Justify.
 - (b) If $x(t)$ is passed through an ideal low pass filter with cut off frequency $\omega_c = 1000\pi$ what is the required Nyquist rate for proper reconstruction?
 - (c) Sketch the spectrum of a sampled signal for over sampling, critical sampling and under sampling.
 - (d) How to reconstruct the signal in over sampling case and what are the requirements on the filter?
 - (e) If $x(t)$ after passing through LPF, with cut of frequency $\omega_c = 1000\pi$ having bandwidth=500Hz is multiplied with sine signal of frequency $\omega_o = 90\text{MHz}$? Sketch the spectrum? What type of sampling is preferred to sample this type of signal? Compute the required sampling frequency?

